Lesson notes · for you

Locks and keys: public-key cryptography

Cyber security · Level 4 · Lesson 2

Time
17 min

Materials
A phone or computer for the lab; A padlock and a small box, or paper to draw them

Learning goals

- Explain the key-sharing problem and how two keys solve it.
- Explain what a digital signature proves.
- Say what the HTTPS padlock does and doesn't prove.

Activity: Padlock post

1. Warm-up (3 min): You and a friend must agree a secret key, but every message passes through a curious stranger. Why is that a problem?
2. Lab (6 min): In the "Locks and keys" lab, pick two primes, make a key pair, lock a word and unlock it. Then sign a message and change one letter.
3. Act it out (5 min): In threes, play the receiver (hands out an open padlock), the sender (locks a note in the box) and the listener (handles the box on the way but can't open it). Swap roles.
4. Wrap-up (3 min): Agree in one sentence what the browser padlock proves, and one thing it doesn't.

One padlock with its key and a small box per group of three, or draw them on paper.

Discussion

- Why is it safe to give your public key to everyone?
- How does a signature stop a fake phone update?
- Why can a scam site still show a padlock?

Check yourself

- Which key locks, which opens, and which signs?

Try it this week
In the lab, make a tiny key pair, lock a word, then sign a message and change one letter after signing. Then, with a friend, open your chat with each other and find where to compare your security codes (in the contact or chat details).

Worksheet

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Name _____ Class _____ Date _____

1 **A page asking for your mobile money PIN shows the HTTPS padlock. Can you trust it because of the padlock?**

- ◯ Yes — the padlock means the site is safe
- ◯ Yes, if the address looks long and official
- ◯ No — it only means the connection is encrypted

2 **Match each idea to what it does.**

Public key	• • • •	Keep it: only it opens and signs
Private key	• • • •	Share it: anyone can lock with it
Digital signature	• • • •	The connection is encrypted
HTTPS padlock	• • • •	Proves who sent it, and that it's unchanged

3 **Someone who copies your public key can lock messages for you, but can't open the ones locked for you.**

- ◯ True
- ◯ False

4 **Put what happens when you open an HTTPS website in order.**

- ☐ The site sends its certificate
- ☐ You open the website's address
- ☐ The padlock appears, and the page arrives encrypted
- ☐ Your browser checks the certificate is for that address

5 **Amal's digital signature on a message checks out. What does that prove?**

- ☐ Nobody else could read it on the way
- ☐ It was signed with Amal's private key
- ☐ What the message says is true
- ☐ Nobody changed it after it was signed

✎ **Where did you rely on public keys today without noticing? Which one surprised you most?**

Check your answers

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1 **No — it only means the connection is encrypted**

Anyone can get a certificate for an address they control, scammers included, so the padlock proves the connection is private, not that the site is honest.

2 **Public key — Share it: anyone can lock with it**
Private key — Keep it: only it opens and signs
Digital signature — Proves who sent it, and that it's unchanged
HTTPS padlock — The connection is encrypted

The two keys do opposite jobs: the public one locks and checks, the private one opens and signs.

3 **True**

Your public key is meant to be copied: it can only lock, never open.

4 **Top to bottom: 2, 1, 4, 3**

The site has to show its certificate before your browser can check it, and the padlock appears only once the check passes and the connection is encrypted.

5 **It was signed with Amal's private key; Nobody changed it after it was signed**

A signature proves who signed and that the message is unchanged.